



Investigation on suppressing vortex-induced vibrations of the rectangular steel box girder for railway cable-stayed bridges by installing wind fairings

Jiahui Dong^b, Lin Huang^b, Haili Liao^{a,b,*}, Qi Wang^{a,b}

^a Key Laboratory for Wind Engineering, Southwest Jiaotong University, Chengdu, 610031, China

^b School of Civil Engineering, Southwest Jiaotong University, Chengdu, 610031, China

ARTICLE INFO

Keywords:

Railway rectangular steel box girder
Vortex-induced vibration (VIV)
Wind fairing
Section model wind tunnel test
Dynamic mesh technology
CFD simulation

ABSTRACT

Severe vortex-induced vibrations (VIVs) may be observed in rectangular steel box girders because of their blunt aerodynamic shape. Herein, the VIV performance and suppression countermeasures of a high-speed railway cable-stayed bridge with a rectangular steel box girder (breadth-to-depth ratio of 6.71) are systematically investigated through experimental and numerical methods. Significant VIVs of the original girder were observed in wind tunnel tests. Then, the dynamic mesh technology is employed to investigate the triggering mechanism. After that, seven types wind fairings are designed. And the flow pattern around the fixed girder installed with them are simulated numerically. Results show that the leading-edge vortex and the after-body vortex shedding cause the deck VIV, and the former plays a major role. The leading-edge vortices can be eliminated by the inclined web with a degree as 16° or 23° . The optimal VIV suppression countermeasure of a DOWNWARD quadrilateral wind fairing with a platform is proposed. After installing the optimal countermeasure, the leading-edge vortex on the bottom surface of the deck disappears, and the vortex on the upper surface becomes smaller and is trapped. Finally, the suppression efficiency of some typical wind fairings is verified through wind tunnel tests.

1. Introduction

Considering the stability and safety of the train operation, the main girder used in long-span high-speed railway bridges needs high stiffness. When box girders are used as the main girder of the long-span high-speed railway cable-stayed bridge, it is necessary to increase the depth of the girder and the angle between the inclined web and the bottom surface to ensure the rigidity. This makes the box-shaped section which finally meets the rigidity requirement a rectangular section with a relatively small breadth-to-depth ratio (less than 7). Although the rectangular cross-section cylinder is simple in geometry, it has become a common research object of bluff body aerodynamics, because its aerodynamic performance has reference significance for the bridge wind resistance. Many researchers have carried out extensive researches on the aerodynamic characteristics and wind-induced response of rectangular cylinders through numerical simulation (Murakami and Mochida, 1995; Yu and Kareem, 1998; Sohankar et al., 1999; Srinivas et al., 2006) and wind tunnel test methods (Namiranian and Gartshore, 1988; McLean and Gartshore, 1992; Norberg, 1993; Matsumoto et al., 2003). And a Benchmark on the Aerodynamics of a Rectangular 5:1 Cylinder

(BARC) was launched in 2008 (Bartoli et al., 2008a, 2008b). A large number of studies (Komatsu and Kobayashi, 1980; Shiraishi and Matsumoto, 1983; Nakamura and Nakashima, 1986; Sun et al., 2009; Bruno et al., 2012) have shown that when the air flows through a bluff body, it will produce flow separations, shear layers, and the phenomenon of vortex generation, merging and shedding. The interaction between the vortex shedding in the wake region and the reattachment of the shear layer will form a complex flow pattern and produce periodic unsteady aerodynamic forces. Therefore, when the rectangular cross-section is applied to the flexible structure, it will lead to VIVs.

VIV is a self-excited oscillation with limited amplitudes which occurs in a low wind speed. Although the VIV response does not always lead to bridge collapse, it seriously impacts the normal operation of the traffic and the fatigue life of components due to its high incidence and large amplitudes. In 2020, VIVs have been observed on Yingwuzhou Yangtze River Bridge in Wuhan and Humen Bridge in Guangdong Province, which not only seriously affects the normal operation of the traffic, but also has a negative impact on public opinion. The VIV of the railway bridge seriously affects the safety of train operations, especially that of the high-speed railway bridge. It should be avoided in design.

* Corresponding author. Key Laboratory for Wind Engineering, Southwest Jiaotong University, Chengdu, 610031, China.

E-mail address: hlliao@swjtu.edu.cn (H. Liao).

<https://doi.org/10.1016/j.jweia.2021.104821>

Received 12 July 2021; Received in revised form 4 October 2021; Accepted 22 October 2021

Available online 7 November 2021

0167-6105/© 2021 Elsevier Ltd. All rights reserved.

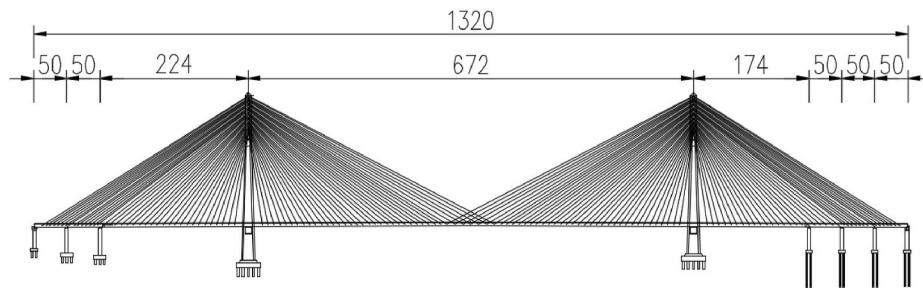


Fig. 1. General layout of the Bianyuzhou Yangtze River Bridge (unit: m)

Typically, wind fairings are attached to the edges of blunt girders to achieve a streamlined shape, and this streamlined shape can improve their aerodynamic performance. But the suppression efficiency of a fairing is in bound up with the original deck section (Nagao et al., 1993). In recent years, the effect of different wind fairings on the aerodynamic stability of different deck sections has been widely discussed. De Mirgnda and Bartoli (2001) designed four different fairings, installed them on the opening cross section deck formed by top slab and side girders, and studied the flutter stability experimentally. Similarly, for the bridge deck with two edge boxes, the optimization efficiency of different triangular wind fairings on the deck flutter has been investigated through experiments and numerical simulations (Sukamta et al., 2008). These studies clearly indicate that an effective fairing can greatly improve the flutter stability of decks with open cross sections. Both Meng et al. (2011) and Li et al. (2018a) investigated the VIV performance and suppression countermeasures of the opening cross section girder with side boxes through experiments. It is demonstrated that for the opening box girder, a smaller wind fairing nose angle can improve the deck VIV performance. Zhang et al. (2020) investigated the VIV performance and suppression countermeasures of a two-box edge girder in detail, and a mini-triangular wind fairing is proposed to suppress the VIV efficiently. Finally, the suppression mechanism of the optimal countermeasure is explained through the CFD method. On the other hand, some authors pay more attention to the efficiency of improving aerodynamic stability of the streamlined trapezoidal box girder by wind fairings. In a study conducted by Wang et al. (2011), it was shown that a box girder with 15° inclined webs has a better flutter stability. Likewise, Larsen and Wall (2012) investigated the VIV performance of a streamlined trapezoidal box girder with three different angles of the inclined web of 26.6° , 19.7° , 14.7° experimentally, and proposed that 14.7° inclined webs could eliminate VIVs of a trapezoidal deck cross section. In addition, the VIV performance of the streamlined trapezoidal box girder

is also sensitive to the angle of the wind fairing nose, and a smaller angle is better (Li et al., 2018b; Li et al., 2018c). For the twin-box girder, Zhou et al. (2018) investigated the nonlinear flutter characteristics of twin-box girder suspension bridges with various slot width ratios and two wind fairing shapes (anti-symmetric and symmetric), and found that the shape of wind fairings could affect the flutter form (symmetric bending-torsional coupled oscillation or anti-symmetric bending-torsional coupled oscillation).

Considering all of this evidence, it seems that researches of rectangular cross-section cylinders are mainly centered on the aerodynamic forces and flow field characteristics, and most literature on the wind fairing is focused on open deck cross sections and streamlined box girders. However, there are few reports to introduce aerodynamic countermeasures to suppress VIVs of rectangular box girders. Although Haque et al. (2016) studied the influence of different triangular wind fairings on rectangular box girders through the CFD method, this study only focuses on the stationary behaviors, not the dynamic performance. Moreover, railings, ballast tubs and some other components are needed for the railway traffic. Thus, compared with the rectangular cross-section cylinders, the rectangular box girder used in railway cable-stayed bridges has worse VIV performance. To ensure the railway operation safety, it is necessary to propose a simple and effective aerodynamic countermeasure to suppress the VIV of the railway cable-stayed bridge with a rectangular box girder. To investigate the aerodynamic characteristics, as everyone knows, visualization of the flow field is an effective method. The CFD and particle image velocimetry methods are commonly adopted to achieve visualization, and the CFD method is more convenient and efficient (Li et al., 2018b). Combined with wind tunnel tests, the CFD method is widely used in the process of exploring the VIV triggering mechanism and proposing aerodynamic suppression countermeasures (Vaz et al., 2016; Xu et al., 2018; Ma et al., 2018).

In this paper, a high-speed railway cable-stayed bridge with a

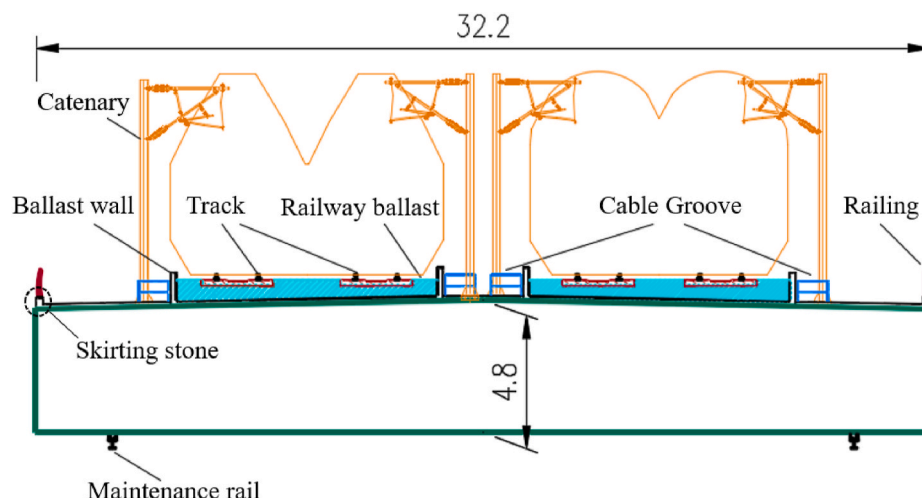


Fig. 2. Cross-section arrangement of the prototype rectangular steel box girder (unit: m).



Fig. 3. 1:50 section model in wind tunnel.

rectangular steel box girder (breadth-to-depth ratio of 6.71) is taken as the engineering background. The main span is 672 m. First, the VIV performance of the original girder is investigated at three attack angles and two damping ratios utilizing a series of 1:50 scale section model wind tunnel tests. To investigate the VIV triggering mechanism of the original rectangular steel box girder, the CFD method and the dynamic mesh technology are employed and the vertical VIV response of the deck under seven wind speeds are calculated. The flow field around the deck under two typical wind speeds are chosen for analysis. After that, seven different types wind fairings are designed, and the flow pattern around the fixed girder installed with different wind fairings is simulated numerically. Based on the simulation results, an effective VIV suppression countermeasure of a DOWNWARD quadrilateral wind fairing with a platform is proposed. Finally, the VIV suppression efficiency of some typical wind fairings is verified through wind tunnel tests.

2. Experimental investigation of VIVs of the rectangular steel box girder

2.1. Engineering background and test setup

The Bianyuzhou Yangtze River Bridge is a long-span high-speed railway cable-stayed bridge located in China. As shown in Fig. 1, the span is arranged as 324 + 672 + 324 m, and 7 auxiliary piers are set at the side span. The design speed of the railway is 350 km/h. A rectangular steel box girder is employed as the main girder to the bridge, and it is supported by two concrete towers with the height of 252 m and 242 m, respectively. The height and width of the girder are 4.8 m and 32.2 m respectively, and the breadth-to-depth ratio of the girder is 6.71. Fig. 2 shows the section layout of the main girder. Based on the bridge location and climatological data, the basic design wind speed of the bridge is 32 m/s.

In order to ensure the safety and stability of the railway operation, wind tunnel tests of the Bianyuzhou Yangtze River Bridge have been conducted at the design stage. The section model wind tunnel tests were carried out in the second test section of XNJD-1 wind tunnel. The cross-section size of the test section is 2.4 m (width) * 2.0 m (height) * 16.0 m (length), and the adjustable wind speed range is 1.0–45.0 m/s. And the turbulence intensity is less than 0.5% in smooth flow. Based on the cross-section size of the main girder and the test section, 1:50 is chosen as the geometrical scale of the section model. The length of the model is 2.095 m, and the length-width ratio is 3.25 which is larger than 2. The blockage ratio is 5.7%, which is not exceed 5%–10% (HAE, 2001). Thereby, blockage correction is not applied. To ensure the enough stiffness of the model and the accuracy of the aerodynamic shape simulation, the model is made of high-quality wood. The railings, maintenance rails, railway ballasts, cable grooves and ballast walls are made of plastic materials (ABS). The catenaries which are arranged with a pitch of 48 m along the direction of the bridge axis are not modeled for the 1:50 section model because of their small effect on aerodynamic performance of the model. Fig. 3 shows the photograph of the section model in wind tunnel. The section model is suspended by eight tension

Table 1
Main parameters of wind tunnel tests.

Properties	Prototype	Scale	Model
Equivalent mass per unit length ($m/kg/m$)	101075	1/50 ²	40.430 (84.701)
Equivalent mass moment of inertia per unit length ($I_m/kg \cdot m^2/m$)	10258256	1/50 ⁴	1.641 (3.438)
Height (H/m)	4.8	1/50	0.096
Length (L/m)	/	/	2.095
Width (B/m)	32.2	1/50	0.644
Vertical frequency (f_h/Hz)	0.2822	/	2.400
Torsional frequency (f_a/Hz)	0.5949	/	4.881
Vertical damping ratio ($\xi_h/\%$)	/	/	0.11, 0.57
Torsional damping ratio ($\xi_a/\%$)	/	/	0.11, 0.48

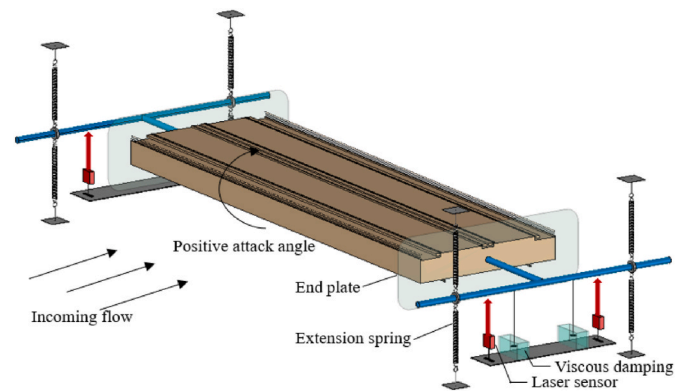


Fig. 4. Diagrammatic sketch of the spring-suspended section model.

springs on the support bracket, which makes it possible for the model to vibrate in vertical and torsion. In order to ensure the two-dimensional flow, two thin end plates are installed. The equivalent mass and mass moment of inertia are simulated by the weight and position of counterweight blocks. In order to adjust the damping ratio of the system, two viscous dampers are employed. Three lasers are set up to obtain the dynamic response. The main dynamic parameters are listed in Table 1. The diagrammatic sketch of the spring suspension system is shown in Fig. 4. The values in parentheses listed in Table 1 are the mass and mass moment of inertia of the section model whose length is 2.095 m. The damping ratios of the system are determined through free-decay tests in wind tunnel tests. The damping of the system is nonlinear and shows amplitude-dependence, especially for the response of torsional mode. Therefore, the damping ratios are determined by the decay time history with an amplitude in the range of the large response of VIVs at this damping ratio.

2.2. Test results of the original girder

The VIVs of the rectangular steel box girder were tested under the

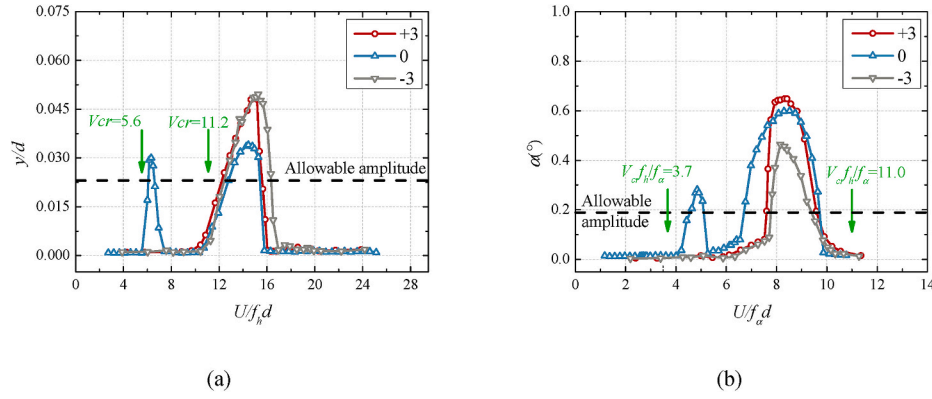


Fig. 5. VIV response of the rectangular steel box girder at $\xi_h = 0.11\%$ ($Sc_h = 7.38$), $\xi_\alpha = 0.11\%$ ($Sc_\alpha = 4.84$): (a) vertical; (b) torsional.

wind speed of 0.5 m/s to 5.8 m/s with an interval of 0.15 m/s (1 m/s corresponding to that of the prototype bridge). The Reynolds number corresponding to experimental investigations is ranged from 3.2×10^3 to 3.7×10^4 . For the highway bridge with a steel box girder, the recommended damping ratio is 0.3% in CMT (2018). The damping ratio for the cable-supported steel bridge are recommended to be 0.36% in HAE (2001). But for the railway bridge with a steel box girder, there is no clear method to estimate the damping ratio before the completion of construction. In order to fully explore the VIV performance of the original bridge section, the VIV responses of the original rectangular steel box girder at attack angles of -3° , 0° and $+3^\circ$ are firstly tested at a low damping ratio which is 0.11%. As shown in Fig. 5, the test results are presented as plots of vertical and torsional VIV response as function of the dimensionless wind speed $U/f_h d$ or $U/f_\alpha d$, where U is the wind speed of the incoming flow, f_h and f_α are the natural vertical and torsional frequency, respectively, and d is the depth of the girder. Sc_h , Sc_α are the Scruton number of the vertical VIV and the torsional VIV, respectively. They can be calculated by $Sc_h = 4\pi m \xi_h / \rho b d$ and $Sc_\alpha = 4\pi I_m \xi_\alpha / \rho b^2 d^2$ (Hansen, 2013), where m is the equivalent mass per unit length, I_m is the equivalent mass moment of inertia per unit length, ρ is the air density which is 1.225 kg/m^3 , b is the width of the rectangular steel box girder, and ξ_h , ξ_α are the damping ratios in vertical and torsion, respectively.

Significant vertical and torsional VIVs are observed at three different angles:

- (1) At the 0° attack angle, two vertical and two torsional lock-in ranges are observed. As Fig. 5(a) shows, the first vertical lock-in range is in the range of $U/f_h d = 5.9\text{--}7.4$, another is in the range of $U/f_h d = 11.1\text{--}15.5$ with a peak dimensionless amplitude of $y/d = 0.033$. Two torsional lock-in ranges are respectively in the range of $U/f_\alpha d = 4.2\text{--}5.3$ and $U/f_\alpha d = 6.7\text{--}9.8$. The maximum

torsional amplitude is 0.60° . Shiraishi and Matsumoto (1983) classified the generation mechanisms of VIVs into three groups and considered that the VIV of the section in group 2 is triggered by two unsteady vortices due to motion. In group 2, the onset critical reduced velocity V_{cr} of VIVs in vertical and torsional was empirically defined for various slenderness ratios $\bar{B}/\bar{D}$ as follows:

$$V_{cr} = U_{cr} / f_h \bar{D} = 1.67 * (1/n) * (\bar{B} / \bar{D}), \text{ for the VIV in vertical} \quad (1)$$

$$= 1.67 * (2/(2n - 1)) * (\bar{B} / \bar{D}), \text{ for the VIV in torsion}$$

where U_{cr} is the onset critical velocity, $n = 1, 2, 3, \dots$, $\bar{B}$ is the effective chord length of cross section, $\bar{D}$ is the effective depth of cross section. $\bar{B}$ and $\bar{D}$ are set as 0.644 m and 0.096 m at the 0° attack angle. Based on the empirical formula (1) for group 2, the vertical V_{cr} of the rectangular steel box girder at 0° attack angle are predicted as 5.6 and 11.2, and the $V_{cr}^*(f_h/f_\alpha)$ for the torsional VIV are 3.7 and 11.0, as shown in Fig. 5. The reduced velocity is defined by the vertical frequency in the reference (Shiraishi and Matsumoto, 1983), which is different in this paper. Thus, the onset critical reduced velocity for the torsional VIV is expressed by $V_{cr}^*(f_h/f_\alpha)$ in this part. Although some discrepancies occur between the predicted torsional onset critical reduced velocity and the experimental result, the high consistency between the predicted V_{cr} in vertical and the test results as well as the existence of the two vertical and two torsional lock-in ranges indicate that the VIV generation mechanisms of the rectangular steel box girder at 0° attack angle probably belongs to group 2, that is, motion-induced vortex vibration. The flow field characteristics of the original section girder and the VIV generation mechanisms will be discussed in detail in section 3.3.

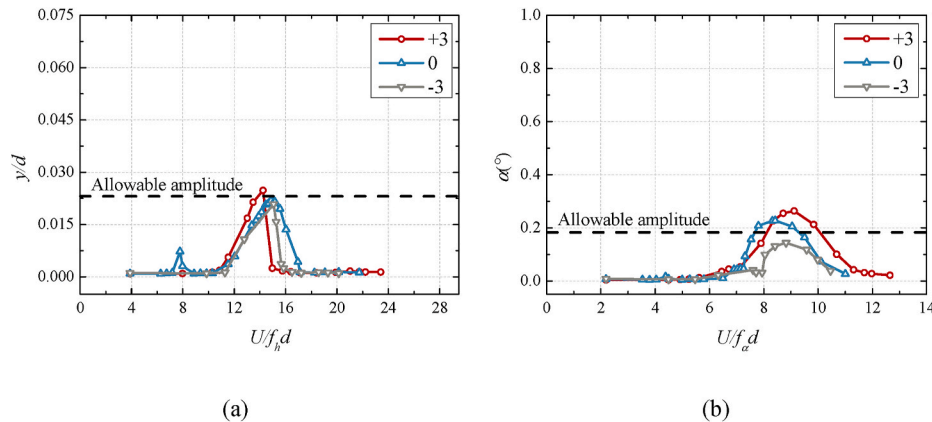


Fig. 6. VIV response of the rectangular steel box girder at $\xi_h = 0.57\%$ ($Sc_h = 38.24$), $\xi_\alpha = 0.48\%$ ($Sc_\alpha = 21.14$): (a) vertical; (b) torsional.

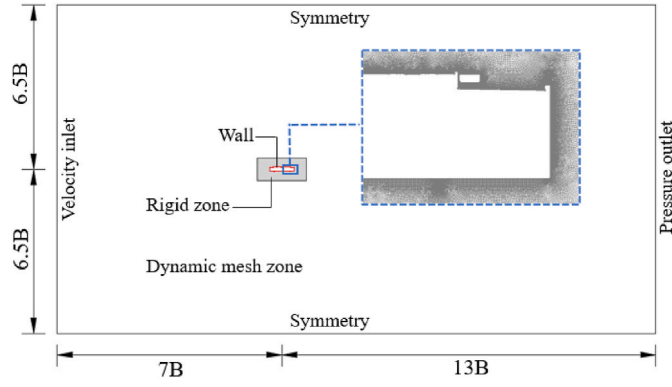


Fig. 7. Computational domain.

- (2) At both $+3^\circ$ and -3° attack angles, one vertical lock-in range in $U/f_h d = 11.1\text{--}16.2$ and one torsional lock-in range in $U/f_a d = 7.7\text{--}9.8$ appear. The maximum dimensionless VIV amplitudes in vertical and torsion are 0.050 and 0.65° , respectively.

Currently, there is no suggestion on the allowable VIV amplitude for railway bridges. In this paper, the allowable VIV amplitude for highway bridges suggested in CMT (2018) is taken as a reference. The vertical and torsional allowable VIV amplitudes can be calculated through following formulas:

$$h_v = \frac{0.04}{f_{vb}} \quad (2)$$

$$\theta_a = \frac{4.56}{B f_{ab}} \quad (3)$$

Where h_v and θ_a are the VIV allowable amplitudes in vertical and torsion respectively; B is the width of the main girder. f_{vb} and f_{ab} are the fundamental vertical and torsional frequencies, respectively. All the parameters in Eq. (2) and Eq. (3) are the real bridge values. Based on Eq. (2) and Eq. (3), the allowable vertical and torsional amplitudes are 141 mm and 0.238° , respectively. Considering the influence of vibrational mode, Wind-Resistant Design Manual for highway bridges in Japan suggests a correction factor for vibrational mode to predict the amplitude of VIVs based on the results of section model wind tunnel tests. The factor is approximately $4/\pi$ (Sato, 2003). That is, the allowable vertical and torsional amplitudes for the section model tests (real bridge conversion value) can be calculated by $h_v/(4/\pi)$ and $\theta_a/(4/\pi)$, respectively. Then, the dimensionless allowable vertical and torsional amplitudes for the section model tests can be obtained, which are 0.023 and 0.187° , respectively. As depicted in Fig. 5, the VIV responses at the 0.11% damping ratio cannot satisfy the allowable VIV amplitudes.

In order to investigate the VIV performance of the original girder at a high damping ratio, further tests are carried out at $\xi_h = 0.57\%$, $\xi_a = 0.48\%$ ($Sc_h = 38.24$, $Sc_a = 21.14$). The test results shown in Fig. 6 indicate that when damping ratios increase to about 0.5%, the amplitudes of vertical and torsional VIVs decrease significantly at three angles of attack. The vertical and torsional responses are both in the critical state of satisfaction and dissatisfaction. Even the vertical response at $+3^\circ$ attack angle and the torsional responses at $+3^\circ$ and 0° attack angles cannot satisfy the allowable amplitude. And for the high-speed railway bridge, the damping ratio in service is unable to accurately obtain before construction. Thus, to ensure the high-speed railway operation safety, effective VIV suppression countermeasures which can suppress the VIVs of the railway rectangular steel box girder in a low damping ratio should be proposed.

Table 2

Computed results of three meshing schemes.

Meshing scheme	Grid number (million)	Drag coefficient	Lift coefficient	Moment coefficient
MS1	0.11	0.911	0.034	0.0013
MS2	0.22	0.902	0.050	0.0008
MS3	0.44	0.901	0.054	0.0006

3. Numerical simulation of the rectangular steel box girder

3.1. Model setup

In this study, in order to have a more intuitive understanding of the VIV triggering mechanism of the original rectangular steel box girder, the CFD software FLUENT was employed to obtain the flow field around the section. The scale of the calculation model is 1:50 of the prototype bridge, which is the same as it of the wind tunnel test. The computational domain is shown in Fig. 7. B is the width of the girder. The length of the computational domain along the flow direction is $20B$ and that across the flow direction is $13B$. Fluid flows from the left boundary (velocity inlet) to the right (pressure outlet). And the upper and lower boundaries are set as symmetry. The bridge girder is set as no-slip wall. Two zones, which are rigid zone and dynamic mesh zone, are set in the computational domain. The rigid zone moves with the bridge girder. It is discretized by unstructured quadrilateral grids. The dynamic mesh zone is discretized by structured quadrilateral grids. The SST $k-\omega$ turbulence model is applied. The SIMPLE algorithm is used for pressure-velocity coupling. About the setting for the spatial discretization method, the least squares cell based method is used for Gradient terms, and the pressure, momentum, turbulent kinetic energy specific dissipation rate are discretized by the second-order upwind scheme.

In the process of the simulation in this section, the bridge girder is fixed first, the flow field around the fixed girder is simulated. When the calculation is relatively stable, the vertical freedom of the girder is released. User-defined functions are used to obtain the lift force on the surface of the girder and bring it to the rigid body motion equation. Then the motion equation is solved by Newmark- β method to obtain the motion state of the girder at the next moment. Dynamic mesh technology is employed to simulate the motion of the girder. The Rigid zone is linked with the girder and moves with it together. The grids in the Dynamic mesh zone are deformed at every time step. And the diffusion-based smoothing method is applied to the deformation of the grids in the Dynamic mesh zone. The mesh motion is governed by the diffusion equation:

$$\nabla \cdot (\gamma \nabla \vec{u}) = 0 \quad (4)$$

where $\vec{u}$ is the mesh displacement velocity, γ is the diffusion coefficient. After Eq. (4) is solved, the node positions are then updated according to:

$$\vec{x}_{new} = \vec{x}_{old} + \vec{u} \Delta t \quad (5)$$

where $\vec{x}_{new}$ and $\vec{x}_{old}$ are the new and the old position of the node, respectively. Δt is the time step.

3.2. Independence study of grid and time step

In the process of the numerical simulation, the calculated results are influenced by the grid size. As shown in Table 2, three meshing schemes with different grid number are created and the computed static coefficients are compared for mesh independence verification. The computed results of MS3 are close to those of MS2, and the results of MS1 are far away from those of MS2 and MS3. Considering the calculation efficiency and accuracy, the meshing scheme MS2 is finally used in the following simulations. The $Y+$ value is less than 4, and the increase ratio is approximately 1.05. The details of the meshing scheme

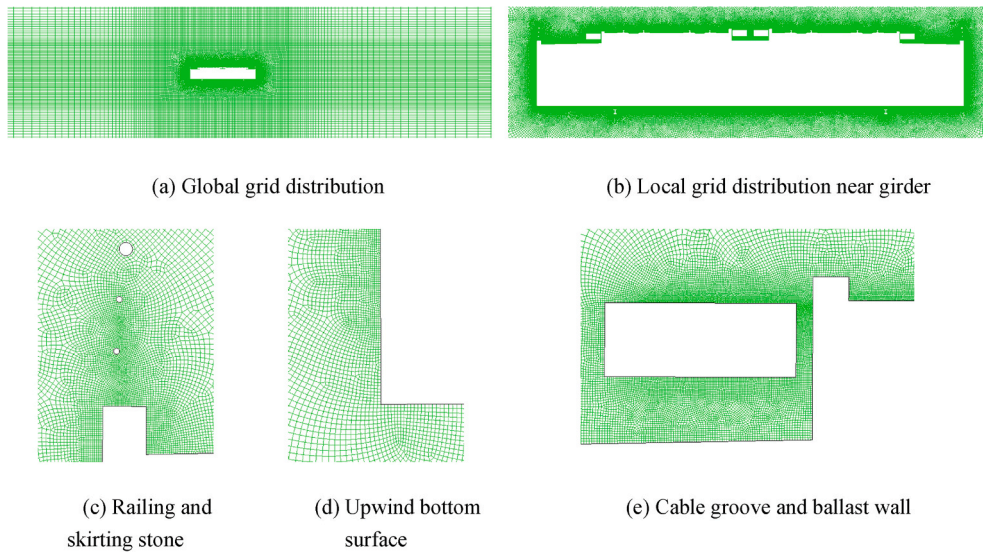


Fig. 8. Grid distributions.

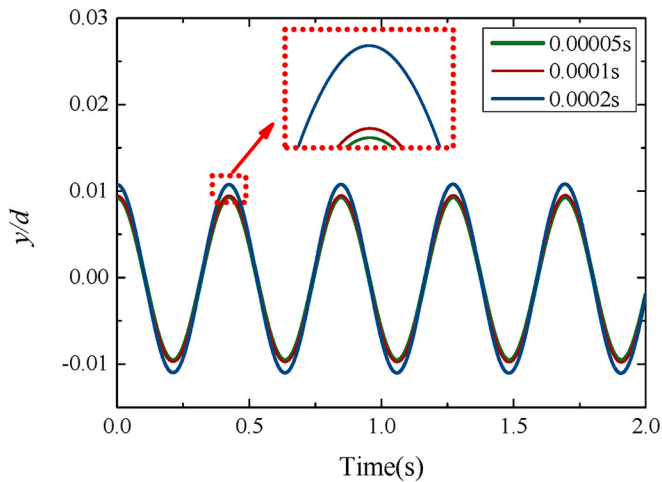
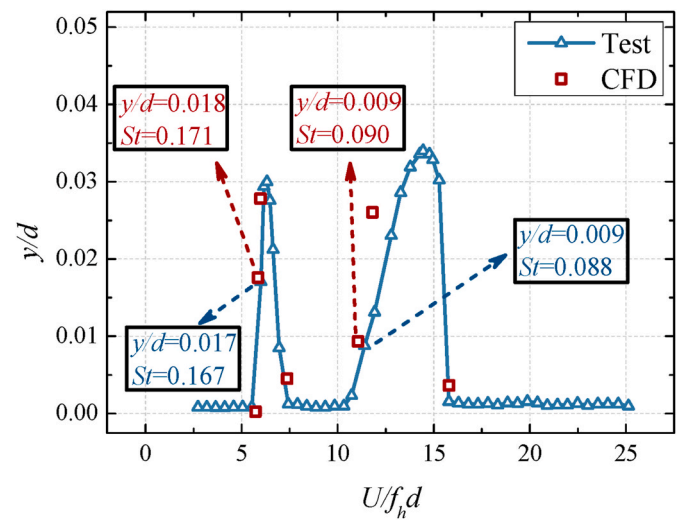


Fig. 9. Time histories of the dimensionless vertical displacement at the three time steps.

MS2 are shown in Fig. 8. And three time steps of 0.00005s, 0.0001s and 0.0002s are chosen to check the independence of the time step on the simulated results. The time histories of the dimensionless vertical displacement at the three time steps are shown in Fig. 9. It can be seen that the amplitudes of the time steps of 0.0001s and 0.00005s are similar, but the result of the time step of 0.0002s is far away from those of the other two. Thus, the time step of 0.0001s is applied to the following numerical simulations.

3.3. Numerical simulation results of the rectangular steel box girder

The dynamic parameters in numerical simulations are set as those in the wind tunnel test. Simulations are only carried out at 0° attack angle. The vertical VIV responses of the original rectangular steel box girder at 0.11% damping ratio are simulated under seven wind speeds ($U_m = 1.32, 1.35, 1.38, 1.70, 2.55, 2.72, 3.64$ m/s). U_m is the wind speed corresponding to the model. The corresponding reduced velocities are $U/f_h d = 5.72, 5.86, 5.99, 7.38, 11.07, 11.80$ and 15.80 . The Reynolds number corresponding to numerical simulations is ranged from 8.4×10^3 to 2.3×10^4 . Since the amplitude-dependence of the damping in vertical mode is not particularly obvious, the damping ratio in CFD simulations are set as a constant (0.11%). The comparison between the

Fig. 10. Simulation and experimental results of vertical VIV amplitudes at 0° attack angle ($\xi_h = 0.11\%$, $Sc_h = 7.38$).

calculated amplitudes and the experimental results is shown in Fig. 10. As can be seen from the graph, the trend of the VIV response obtained in the numerical simulation approach that measured in the wind tunnel test. The vertical response frequencies at $U/f_h d = 5.86, 5.99, 7.38, 11.07, 11.80$ and 15.80 are 2.37 Hz, 2.46 Hz, 2.34 Hz, 2.31 Hz, 2.35 Hz, 2.42 Hz. They are approximately consistent to the natural frequency of the model in wind tunnel which is 2.40 Hz. The Strouhal number can be calculated by $St = f_h d/U$, where St is the Strouhal number, f_h is the natural vertical frequency, d is the height of the girder and U is the mean wind speed of the incoming flow. As shown in Fig. 10, the onset wind speeds of the two lock-in ranges agree well with the test results, and the Strouhal number is in good agreement with that calculated through test results. In this section, the purpose of employing CFD is to explore the triggering mechanism of VIVs. Thus, the good agreement of the Strouhal number makes it possible to consider that the flow field characteristics obtained in the numerical simulation are reliable. The instantaneous vorticity magnitude and pressure distribution around the original rectangular steel box girder over one vibration cycle at $U/f_h d = 5.86$ and $U/f_h d = 11.07$ are shown in Fig. 11 and Fig. 13. The time histories of C_L and the dimensionless vertical displacement under $U/f_h d = 5.86$ and $U/f_h d$

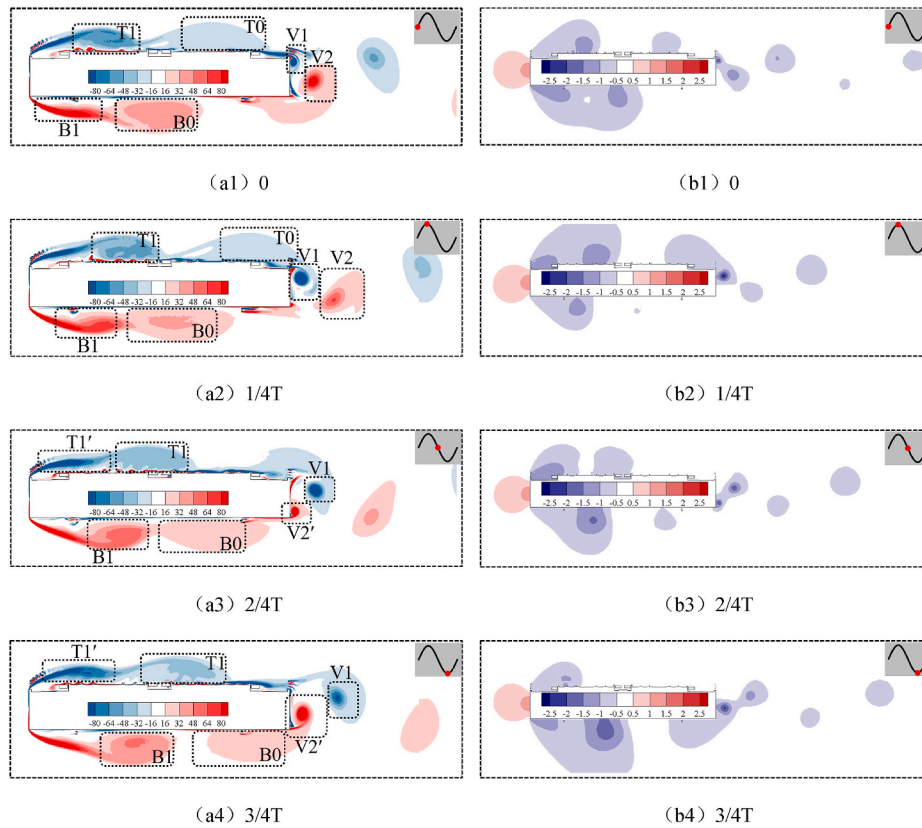


Fig. 11. The instantaneous vorticity magnitude and pressure distribution around the original rectangular steel box girder over one vibration cycle at $U/f_h d = 5.86$.

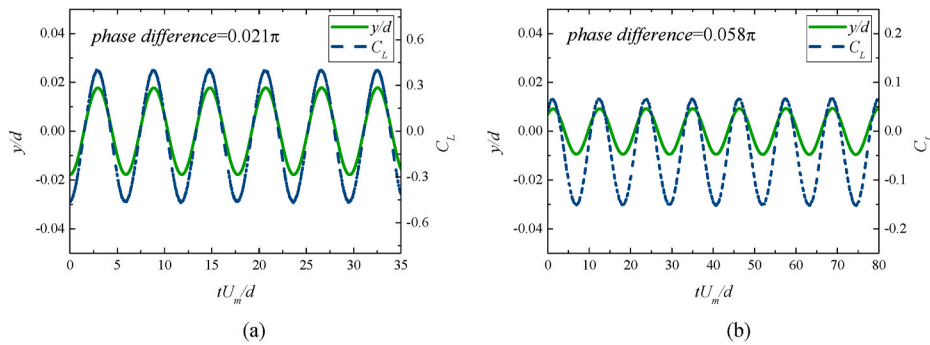


Fig. 12. Time histories of C_L and the dimensionless vertical displacement at (a) $U/f_h d = 5.86$ and (b) $U/f_h d = 11.07$.

$= 11.07$ are shown in Fig. 12. It can be seen that the phase difference between C_L and y/d is very small whether at $U/f_h d = 5.86$ or $U/f_h d = 11.07$. These indicate that the aerodynamic lift force does positive work at both the two reduced wind velocities. Since the phase difference is very small, it is ignored in the description for Figs. 11 and 13 later.

In Fig. 11(a1), as the deck moves up, the B1 vortex forms, the T1 vortex sheds from the leading edge. The V1 vortex forms in the wake region, and the V2 one moves away. The T0 and B0 vortices shedding in the last cycle move downstream along the deck surface. At this moment, the deck experiences the mean lift force, as shown in Fig. 11(b1). Then, when the deck moves to the top, the B1 vortex becomes mature and the T1 one completes vortex shedding and moves downstream along the deck surface. Meanwhile, the V1 vortex completes vortex shedding and starts merging with the vortex T0. The B0 vortex continues to move downstream. The lift force of the main girder reaches the maximum value at this time. After that, as the deck moves down, a new vortex called T1' forms at the leading-edge, and the B1 vortex sheds and starts moving downstream. The V2' vortex sheds in the wake region and the

V1 one moves away. As can be seen in Fig. 11(a4), the deck moves to the bottom, the B1 vortex and the V2' one shed completely, and the T1' vortex becomes mature. The lift force of the main girder reaches the minimum value at this time. Then the deck returns again to Fig. 11(a1) and continues to the next cycle.

As shown in Fig. 13, in the second lock-in range, the Karman vortex street is not obvious in the wake region and the wake shows fish-tailing, which is different from the flow pattern in the first lock-in range. When the section moves from the balance position to the highest, the T1 vortex moves along the downwind direction, and the height of the B1 vortex gradually increases. In this process, the pressure difference between the upper and lower surfaces of the beam increases gradually, and the lift force direction is consistent with the movement direction. Until the girder moves to the highest, the lift force of the main girder reaches the maximum. Then, the girder moves downward, the T1 vortex becomes larger. At the lower side, the B1 vortex is mature and moves along the downwind direction. When the girder moves below the balance position, the direction of resultant lift force is consistent with the direction of

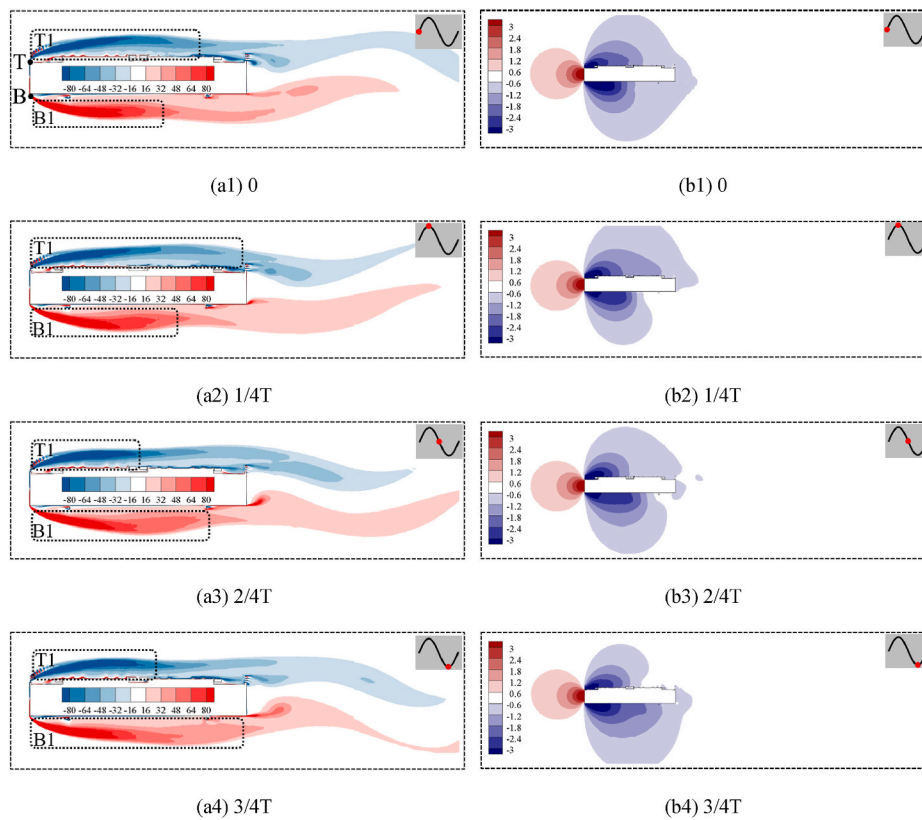


Fig. 13. The instantaneous vorticity magnitude and pressure distribution around the original rectangular steel box girder over one vibration cycle at $U/f_h d = 11.07$.

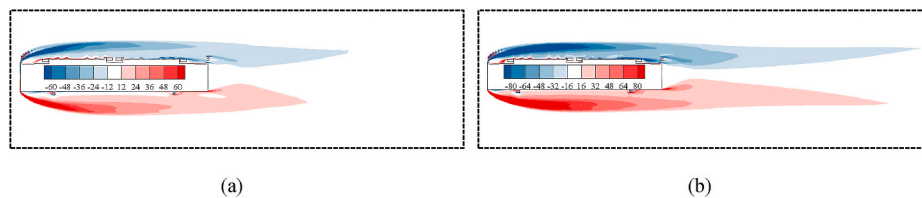


Fig. 14. The vorticity magnitude around the static girder at (a) $U/f_h d = 5.86$ and (b) $U/f_h d = 11.07$.

movement. The absolute value of the lift force increases gradually and reaches the maximum when the girder moves to the lowest. Then the deck returns again to Fig. 13(a1) and continues to the next cycle. The two leading-edge vortices move along the downwind direction alternately and oscillate like fish-tailing in the wake region, which also contributes to the periodic vibration of the beam.

Through the above analysis, we can find that the flow patterns around the main girder are different in the two VIV lock-in regions. The wind flows through the main girder and separates at the leading edge, generating two vortices. In the first lock-in range, the two leading-edge vortices shed alternately with the vibration of the main girder, which exerts a periodic force on the main girder. The shedding vortices move along the downwind direction on the upper and lower surface of the beam and merges with the wake vortex at the trailing edge. And there is a typical Karman vortex shedding in the wake region. In the second lock-in range, the two leading-edge vortices do not shed. The two leading-edge vortices only oscillate and move along the downwind direction alternately, and oscillate like fish-tailing in the wake region. However, when the main girder is fixed, the vortex shedding in the wake region under $U/f_h d = 5.86$ and $U/f_h d = 11.07$ is apparently weak, as shown in Fig. 14. And the weak fluctuation of the lift coefficient of the static girder can also confirm it. The fluctuating lift coefficients are 0.0006 under $U/f_h d = 5.86$ and 0.0007 under $U/f_h d = 11.07$. The similar flow pattern is

also found around the rectangular cylinder with breadth-to-depth ratios between 6.0 and 7.0 by Shimada and Ishihara (2002) through numerical methods. It is indicated that the shedding or oscillating of the after-body vortex of the rectangular steel box girder is caused by the motion and is influenced by the leading-edge vortex.

Thus, for the original rectangular steel box girder with a breadth-to-depth ratio of 6.71, the triggering mechanism of VIV should be that, a small disturbance causes the leading-edge vortex to periodically shed or oscillate, then, the periodically shedding or oscillating vortices move along the downwind direction and merge with the after-body vortices in the wake region with the same frequency. The pressure distribution in Fig. 13 indicates that the fluctuation of the lift force during the vibration of the main girder is mainly caused by the leading-edge vortex oscillating on the upper and lower surfaces of the main girder, and the wake also contributes to it, but it is relatively small. The same conclusion can be drawn from the pressure distribution in Fig. 11. Therefore, the leading-edge vortex is considered to be the main cause of VIVs of the original section. It is necessary to find an aerodynamic countermeasure to reduce or even eliminate the leading-edge vortex to suppress VIVs of the main girder in the further study.

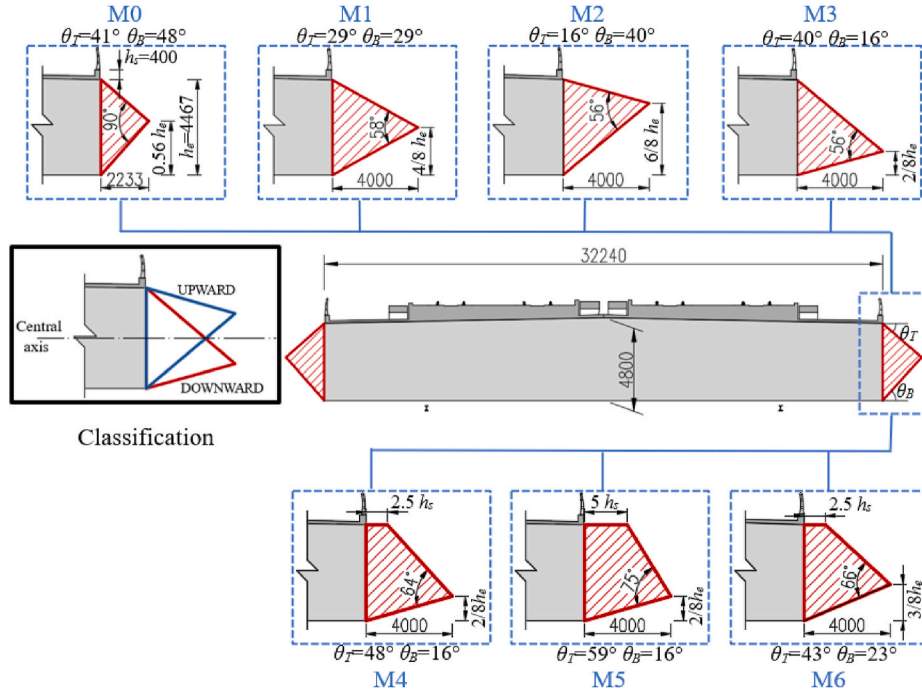


Fig. 15. Classification and configuration of wind fairings (units: mm).

Table 3

Computed results of three meshing schemes for the section with M1 wind fairing.

Meshing scheme	Grid number (million)	S_i
S1	0.12	0.1612
S2	0.24	0.1643
S3	0.48	0.1648

4. The aerodynamic shape selection of wind fairings based on numerical simulations

At present, the CFD method is the most convenient and efficient method to realize visualization. In the process of looking for aerodynamic countermeasures, obtaining the flow field around the girder installed with different countermeasures through the CFD method can guide the optimization of aerodynamic countermeasure more visually.

4.1. Triangular wind fairing

As effective aerodynamic measures, wind fairings are widely installed on bridges to improve their aerodynamic performance (Zhou et al., 2020). And it is widely known that a smaller angle makes the fairing have higher suppression efficiency (Meng et al., 2011; Li et al., 2018a; Li et al., 2018b; Li et al., 2018c). In this section, four types triangular wind fairings (M0~M3) are designed first, as shown in Fig. 15. h_e in Fig. 15 is the height of the girder edge, and h_s is the height of the skirting stone. The dimension of the fairing corresponds to that of the prototype bridge. In this study, the fairing whose nose is above the central axis is called UPWARD fairing, and the fairing whose nose is below the central axis is called DOWNWARD fairing, as classified in Fig. 15. M0 fairing with $\theta_T = 41^\circ$ and $\theta_B = 48^\circ$ and M1 fairing with $\theta_T = \theta_B = 29^\circ$ are almost symmetrical about the central axis, and the angle of their nose are 90° and 58° , respectively. Considering the asymmetry of the upper and lower surface of the girder, UPWARD and DOWNWARD fairing are designed, such as M2 and M3 fairing. The nose angles of them are 56° . The wind fairing M1 are selected for the mesh independence verification for the section installed with wind fairings. As shown in Table 3, three meshing schemes with different grid number are created

and the computed Strouhal number are compared for mesh independence verification. The computed result of S3 are close to that of S2, and the result of S1 is far away from those of S2 and S3. Considering the calculation efficiency and accuracy, the meshing scheme S2 is finally used in the following simulations. The mesh scheme for the section with wind fairings are similar to that for the section without the wind fairing. Thus, the same time step of 0.0001s is used in the following simulations.

The flow field around the fixed girder installed with M0~M3 wind fairing is simulated at $U_m = 3$ m/s by CFD methods, results are shown in Fig. 16. The attack angle is 0° . The Reynolds number corresponding to the numerical simulations is 1.9×10^4 . The C_L of the original rectangular section at 0° attack angle do not fluctuate with time, and there is no superior frequency in the time history. Therefore, the instantaneous velocity field of it is presented at only one moment.

Compared with the original section, the two leading edge vortices of the main girder section with M0 ~ M3 fairing are reduced in a certain extent. As shown in Fig. 16(b), although θ_T and θ_B of M0 wind fairing are not small, the leading-edge vortices on both upper and lower surfaces are reduced after installing M0 wind fairing. As the nose angle of the wind fairing becomes smaller, for example, after the installation of M1 wind fairing with 58° nose angle, leading-edge vortices are further reduced. There is only a small bubble at the leading edge of the lower surface. But the leading-edge vortex on the upper surface is still higher than the height of the ballast wall. This still leads to a periodic reattachment on the upper surface. The after-body flow field of the section installed with wind fairings is compared with that of the original section. It is found that installing wind fairings makes the after-body flow field change from almost symmetrical to periodic vortex shedding. Because of the existence of the inclined web of the fairing, the periodic vortex-induced force produced by the vortex shedding in the wake region can be applied to the girder more effectively. The participation of the after-body vortex shedding becomes higher after installing wind fairings. Therefore, the installation of an inappropriate wind fairing may weaken the VIV performance of the railway rectangular girder.

After the installation of M2 fairing with $\theta_T = 16^\circ$, the leading-edge vortex on the upper surface is greatly reduced. It makes the vortex smaller than the height of the ballast wall. Then, the vortex is trapped and cannot move. Therefore, the flow will not reattach on the upper

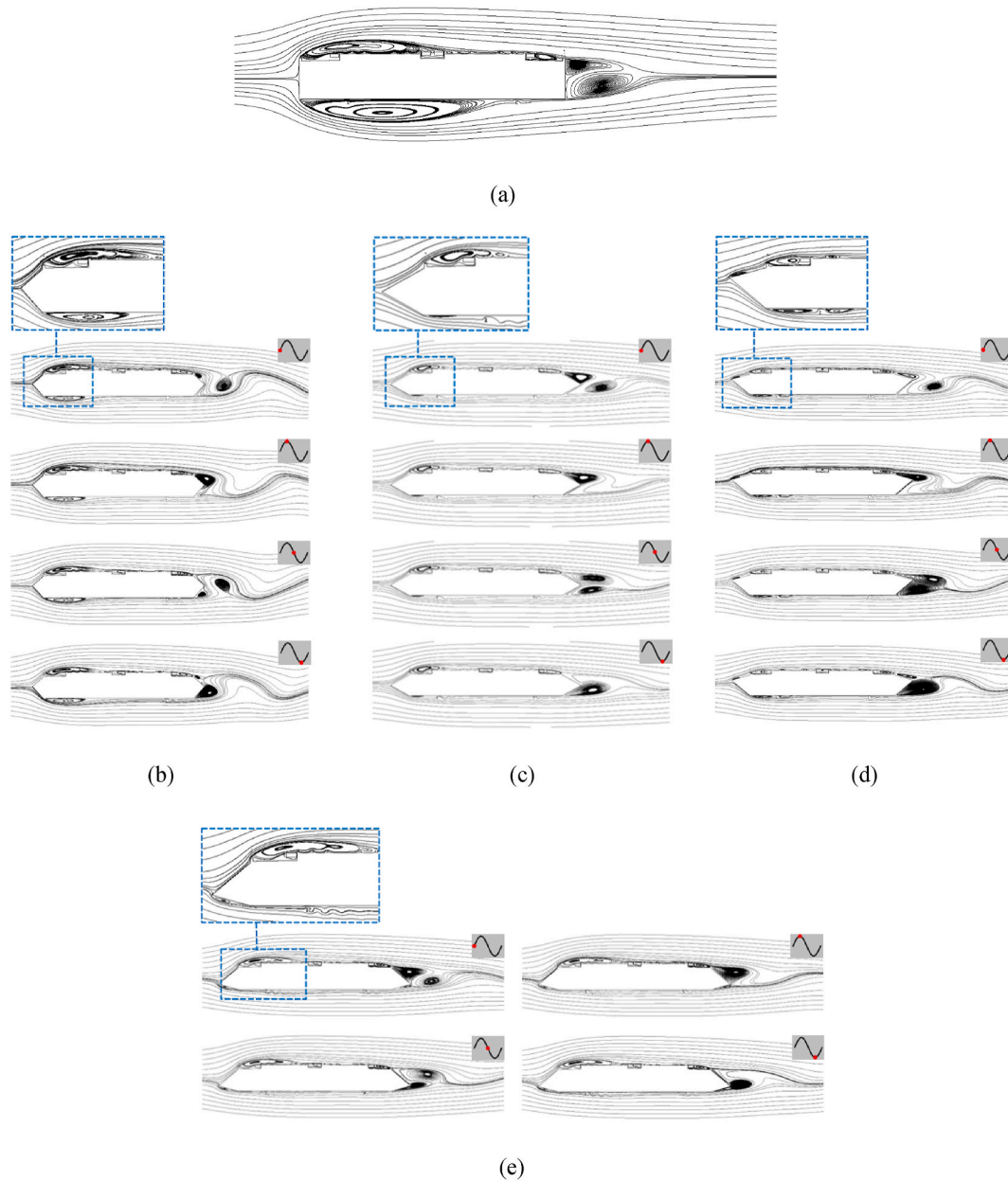


Fig. 16. Instantaneous velocity fields for the section of (a) the original rectangular box girder and the girder installed with (b) M0 fairing, (c) M1 fairing, (d) M2 fairing and (e) M3 fairing along one lift cycle at 0° attack angle.

surface alternately. However, the leading-edge vortex on the lower surface of the girder still exists because of the large θ_B which is 40° . Different from M2 fairing, M3 fairing has a larger θ_T which is 40° and a smaller θ_B which is 16° . It can be seen from Fig. 16 (e) that the leading-edge vortex on the upper surface is larger than the height of the ballast box, which lead to the flow periodical reattachment on the upper surface of the girder. And the small θ_B makes the disappearance of the bottom leading-edge vortex. It is thus clear that smaller θ_T and θ_B can reduce or even eliminate the leading-edge vortices. However, due to the limitation of the wind fairing length, the triangular fairing cannot keep both θ_T and θ_B at a small value, such as 16° . If we simply extend the length of the symmetrical triangular wind fairing to make both θ_T and θ_B reach 16° , an 8 m long wind fairing will be obtained. This is obviously uneconomic and will increase the installation difficulty. Due to the existence of the secondary dead load, aerodynamic shapes of the upper and lower surfaces of the railway rectangular steel box girder are not the same. Therefore, the upper and lower inclined web of the wind fairing can be

considered separately. The leading-edge vortex at the bottom of the girder can be eliminated by a small θ_B . Kubo et al. (1993) proposed a truncated fairing with multi separation points to improve the aerodynamic instability of the girder. The concept to the effect of this kind of wind fairing on improving the aerodynamic instability is that letting the flow separate at the first separation point and reattach on the slope surface can weaken the strength of the separation flow from the second separation point. In this paper, the large leading-edge vortex on the upper surface is generated due to the flow separation at the right-angle skirting stone. Referring to the concept by Kubo et al. (1993), if the flow can separate before reaching the stone, the leading-edge vortex on the upper surface may be greatly reduced. Thus, a new type of wind fairing is obtained by adding a platform at the top of the triangular wind fairing. This fairing type is called DOWNWARD quadrilateral wind fairing with a platform in this paper. According to this, M4~M6 wind fairings are designed, as shown in Fig. 15.

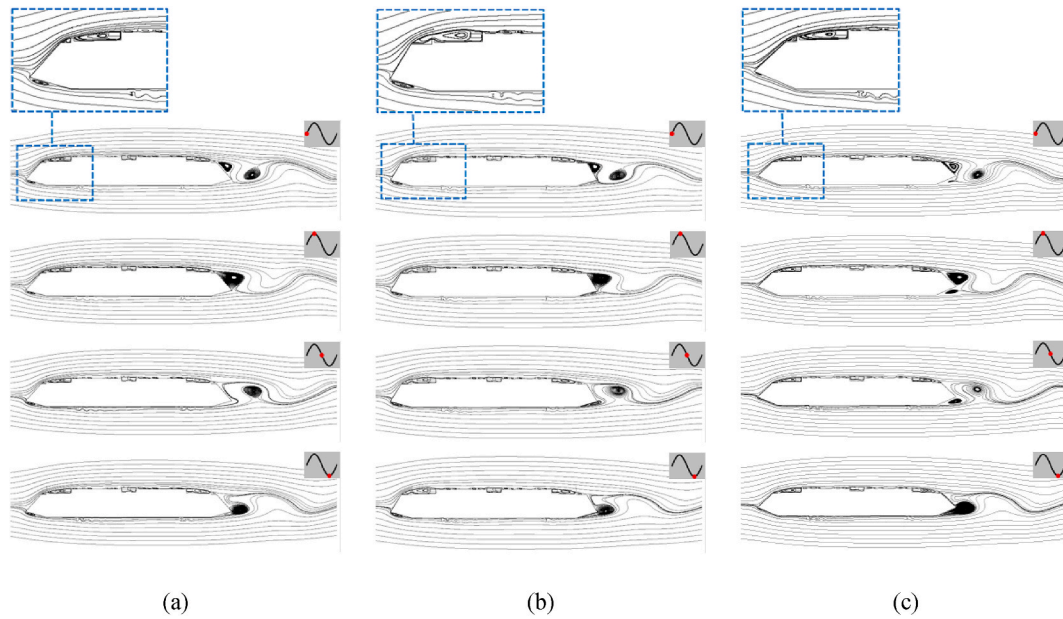


Fig. 17. Instantaneous velocity fields for the section installed with (a) M4 fairing, (b) M5 fairing and (c) M6 fairing along one lift cycle at 0° attack angle.

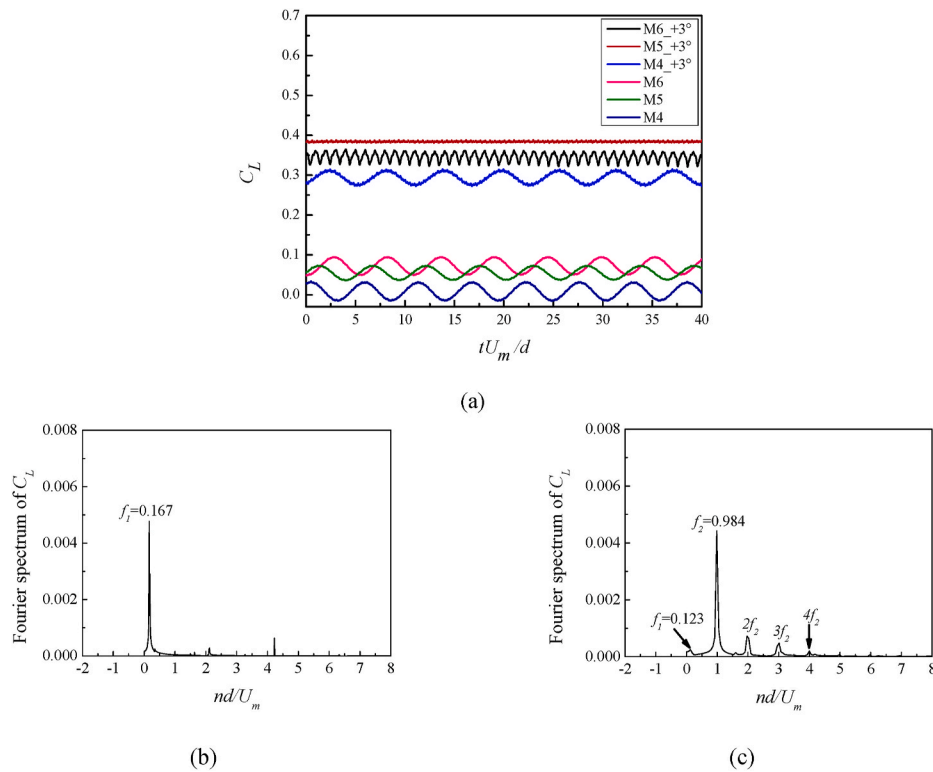


Fig. 18. Fluctuations of the lift coefficient (C_L) of the section with M4 ~ M6 fairing at $+3^\circ$ attack angle: (a) Time history; (b) Fourier spectrum of C_L of the section with M4 fairing; (c) Fourier spectrum of C_L of the section with M6 fairing.

4.2. DOWNWARD quadrilateral wind fairing with a platform

As shown in Fig. 15, the lengths of M4~M6 wind fairing which are 4 m, are the same as those of M1~M3 wind fairing. Platforms with lengths of 1 m, 2 m and 1 m are set at the upper side of M4-M6 fairing. The θ_B of M4 and M5 wind fairing are the same, it is 16° . And the θ_B of M6 wind fairing is 23° . The flow field around the section with M4 ~ M6 fairing at 0° attack angle are simulated first, results are shown in Fig. 17.

The velocity field shows that the leading-edge vortices at the bottom

of the girder are eliminated by installing these three fairings, only three small bubbles are generated under the nose of the leading-edge fairing. And the bubble under M6 fairing is smaller than the other two bubbles. Although the θ_T of M4, M5 and M6 fairing which are 48° , 59° and 43° respectively, are larger than those of M1, M2 and M3 fairing, the existence of the platform makes the flow separate at the intersection of the upper inclined web and the platform along the upper inclined web, bypass the right-angle skirting stone and generate a small bubble after the skirting stone. The small bubble is trapped between the stone and the

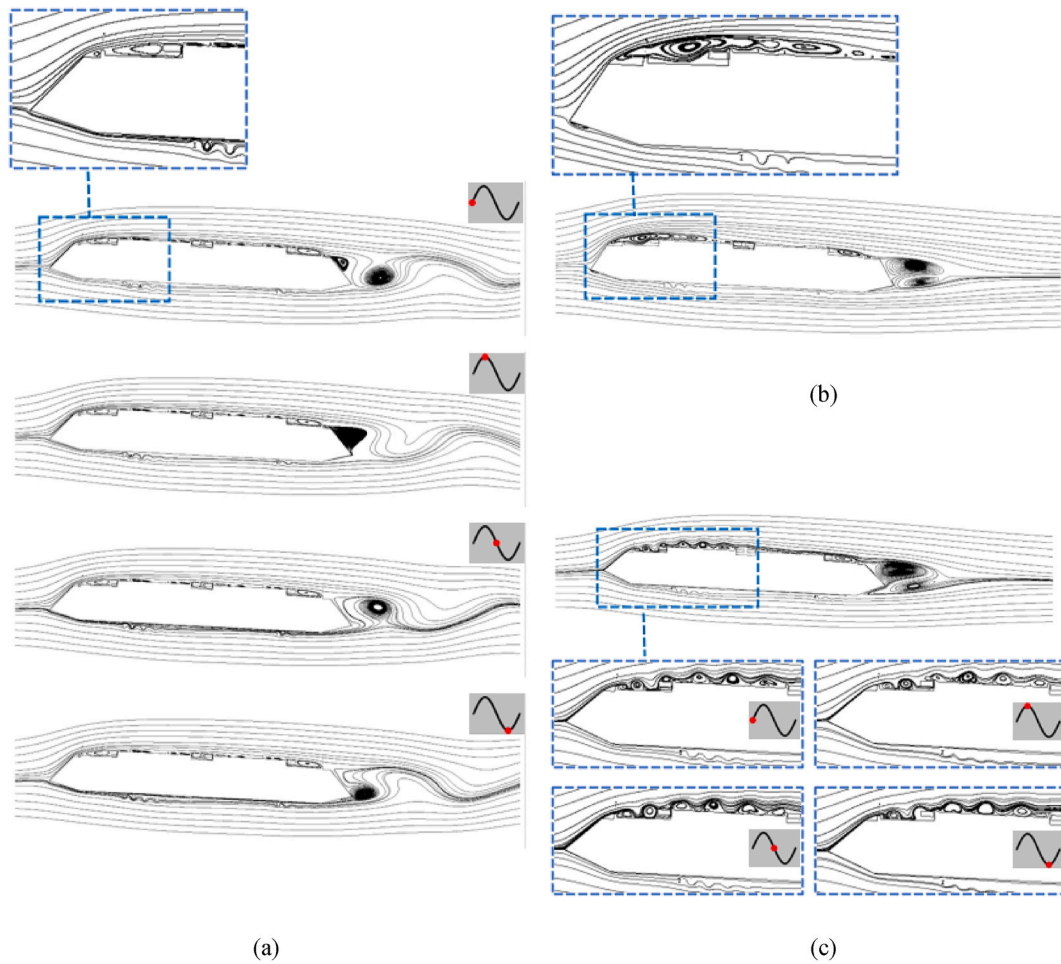


Fig. 19. Instantaneous velocity fields for the section installed with (a) M4 fairing, (b) M5 fairing and (c) M6 fairing along one lift cycle at $+3^\circ$ attack angle.

ballast wall, which makes the flow no longer alternately reattach on the upper surface. It is thus clear that the installation of M4~M6 wind fairing can effectively reduce or even eliminate the leading-edge vortices, which will greatly reduce the VIV response of the girder.

In order to further compare the VIV suppression efficiency of M4~M6 wind fairing, the flow fields around the section with M4 ~M6 wind fairing at $+3^\circ$ attack angle are simulated, results are shown in

Fig. 18 and Fig. 19.

Fig. 18(a) shows that, at $+3^\circ$ attack angle, the mean lift coefficients of the section installed with M4, M5 and M6 wind fairing are significantly larger than those at 0° attack angle. This indicates that, compared with that at 0° attack angle, the effect of the after-body vortex shedding on VIV is enhanced at $+3^\circ$ attack angle. At $+3^\circ$ attack angle, the C_L of the section with M4 and M6 wind fairing fluctuates, but the C_L of the section with M5 wind fairing does not fluctuate with time. The Fourier spectrum of C_L of the section with M4 and M6 wind fairing are depicted in Fig. 18 (b) and (c). For the section with M4 wind fairing, the time history of its C_L has only one main frequency ($nd/U_m = 0.167$), as Fig. 18(b) shows. n is the natural frequency of the fluctuation of C_L , which is calculated through fast Fourier transform. d is the height of the girder. U_m is the wind speed corresponding to the model. The flow field along one lift cycle corresponding to this frequency is shown in Fig. 19(a). The instantaneous velocity field shows that there is no leading-edge vortex existed on the bottom surface of the section, the leading-edge vortex on the upper surface is trapped, and only vortices exist in the wake region. It can be found that the fluctuation of C_L is caused by the periodic vortex shedding in the wake region. Fig. 19(b) shows the instantaneous velocity field at one moment around the section with M5 wind fairing. Since the C_L of the section with M5 wind fairing does not fluctuate with time, the instantaneous velocity field of it is presented at only one moment. Due to the same small θ_B as M4 wind fairing, for the girder installed with M5 wind fairing, the leading-edge vortex on bottom surface does not exist. Compared with the M4 one, after installing M5 wind fairing, no vortex sheds in the wake region, but the leading-edge vortex on the upper surface is larger, and it is higher than the ballast wall. Fig. 18 (c) shows

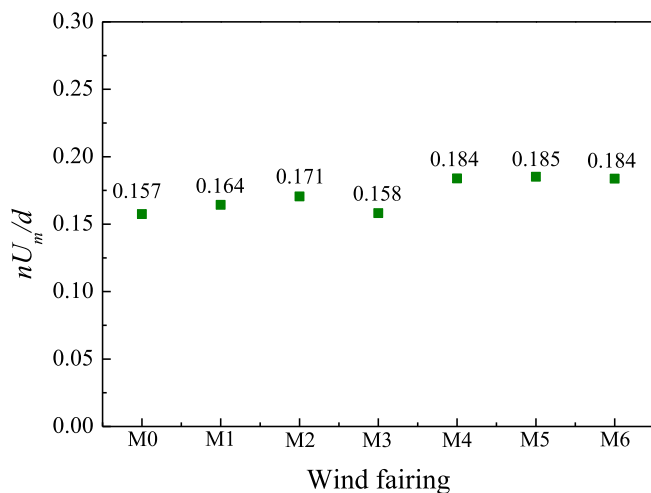


Fig. 20. The reduced frequencies of C_L of the section installed with M0-M6 wind fairings at 0° attack angle.

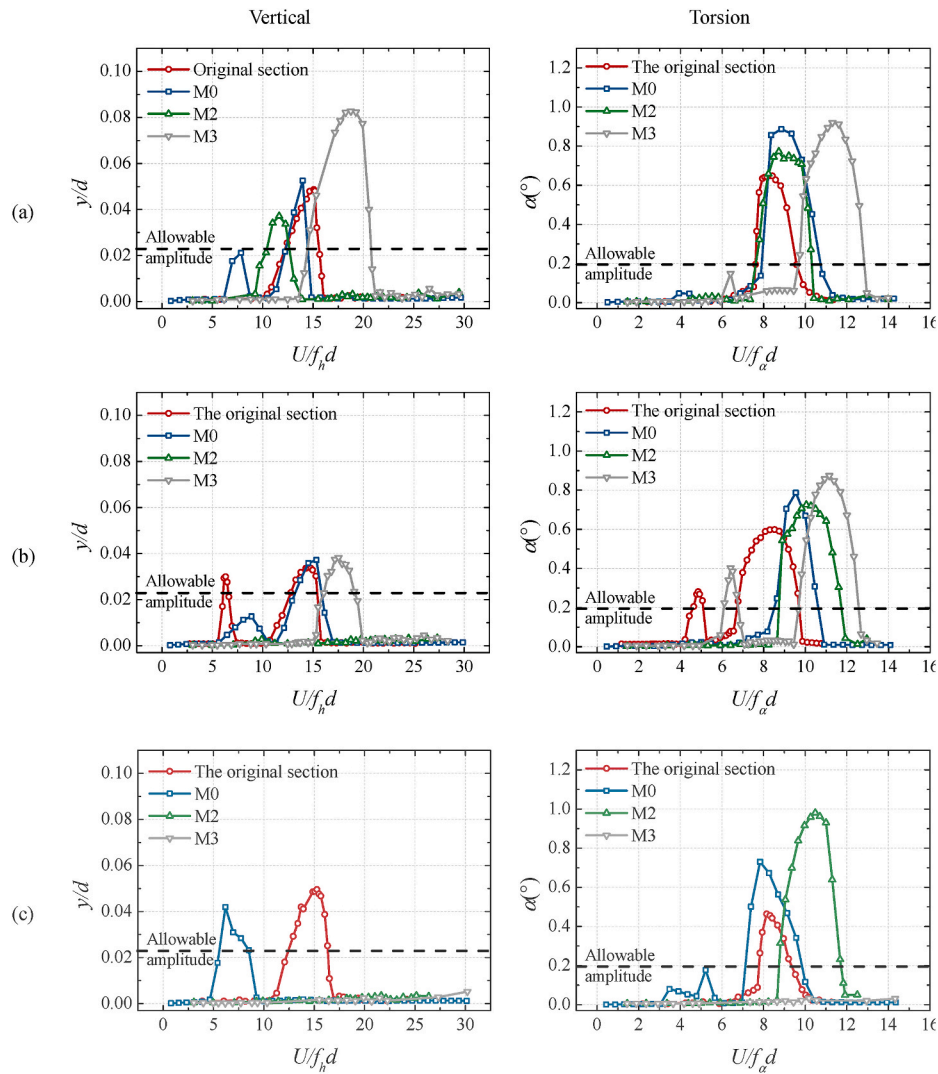


Fig. 21. VIV response of the girder installed with M0, M2 and M3 wind fairing at $\xi_h = 0.11\%$ ($Sc_h = 7.38$), $\xi_a = 0.11\%$ ($Sc_a = 4.84$): (a) at $+3^\circ$ attack angle, (b) at 0° attack angle and (c) at -3° attack angle; the dimensionless allowable vertical and torsional amplitudes are 0.023 and 0.187° , respectively.

five peaks at $nd/U_m = 0.123, 0.984, 1.969, 3.015$ and 3.999 . The flow field around the section with M6 fairing at $+3^\circ$ attack angle along one lift cycle is under $nd/U_m = 0.984$, as shown in Fig. 19 (c). From the flow field, it can be seen that the higher reduced frequencies ($nd/U_m = 0.984, 1.969, 3.015$ and 3.999) are caused by the shedding vortices generated at the upper leading edge. The fluctuation corresponds to $nd/U_m = 0.123$, is apparently weak compared to that of the section with M4 fairing. It is similar to that of the section installed with M5 fairing. At $+3^\circ$ attack angle, both the existence of the high frequency leading-edge vortex-induced force and the weak after-body vortex shedding make the VIV suppression efficiency of M6 wind fairing the best among the three wind fairings.

Based on the above simulation results at 0° and $+3^\circ$ attack angle, the M6 wind fairing is considered to be the optimal one.

The reduced frequencies of C_L of the section installed with M0-M6 wind fairing at 0° attack angle are shown in Fig. 20. It can be seen that (1) nd/U_m increases with the increase of the length of the wind fairing (M0 and M1 wind fairing); (2) nd/U_m increases with the increase of the nose position of the wind fairing (M1, M2 and M3 wind fairing); (3) adding platforms at the top of the triangular wind fairing makes nd/U_m increase (M3, M4 and M5 wind fairing). The reason may be that any of the above changes makes the development space of the upper trailing-edge vortex smaller. And the smaller θ_T of the triangular wind fairing

and the existence of the platform of the quadrilateral one make the horizontal velocity of the shedding vortex on the upper surface become larger. Thus, nd/U_m increases.

5. Experimental investigation on VIV suppression efficiency of wind fairings

In previous section, the influence of wind fairings on the flow field around the fixed girder was qualitatively analyzed by the CFD method, and the VIV suppression efficiency of the wind fairing was preliminarily discussed. To investigate the actual suppression efficiency of the wind fairing and verify the accuracy of the CFD qualitative analysis, M0, M2, M3, M4, M5, M6 wind fairing are chosen, and the vertical and torsional VIV amplitudes of the girder installed with them are tested at $+3^\circ, 0^\circ$ and -3° attack angle. The damping ratio is set as $\xi_h = 0.11\%$, $\xi_a = 0.11\%$. Test results are shown in Fig. 21 and Fig. 22.

As Fig. 21 shows, installing M0 fairing cannot reduce the VIV, and even increase the torsional amplitude at three attack angles. This result supports the viewpoint that the installation of an inappropriate wind fairing may weaken the VIV performance of the railway rectangular girder in section 4. After the installation of M2 fairing, the vertical VIV response is disappeared at 0° and -3° attack angles, and at $+3^\circ$ attack angle, the maximum dimensionless vertical amplitude is reduced to

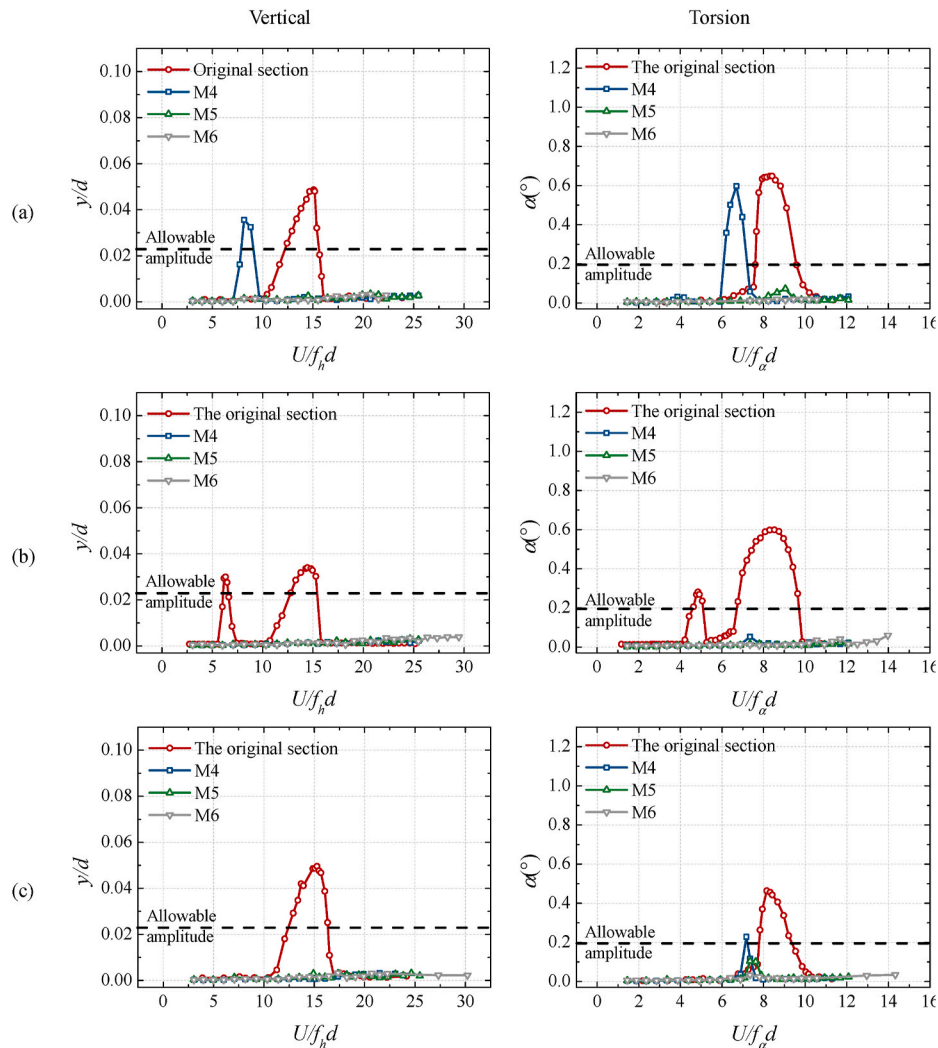


Fig. 22. VIV response of the girder installed with M4, M5 and M6 wind fairing at $\xi_h = 0.11\%$ ($Sc_h = 7.38$), $\xi_\alpha = 0.11\%$ ($Sc_\alpha = 4.84$): (a) at $+3^\circ$ attack angle, (b) at 0° attack angle and (c) at -3° attack angle; the dimensionless allowable vertical and torsional amplitudes are 0.023 and 0.187° , respectively.

0.037 , but the torsional amplitudes at three attack angles increase. Especially at -3° attack angle, the torsional amplitude reaches 0.980° . M3 fairing eliminates the VIVs at -3° attack angle, but weakens the VIV performance at the other two attack angles. Maximum dimensionless amplitudes occur at $+3^\circ$ attack angle, which are 0.083 and 0.921° in vertical and torsion, respectively.

Fig. 22 shows that the VIV performance can be effectively improved by installing M4 fairing, and the VIV amplitudes at three attack angles are remarkable reduced, only the VIV amplitude at $+3^\circ$ attack angle cannot satisfy the Chinese design specification CMT (2018). The Strouhal number of the section with M4 fairing at $+3^\circ$ attack angle is 0.169 which is calculated from the test result in Fig. 22(a). It is very close to the reduced frequency ($nd/U_m = 0.167$) which is calculated based on the simulation result in Fig. 18(b). All of these illustrate that, at $+3^\circ$ attack angle, the VIVs of the section with M4 fairing is mainly caused by the vortex shedding in wake region. Installing M6 fairing can eliminate VIVs at all three attack angles, and if M5 fairing is installed, the VIV performance will also be remarkable improved, only small torsional amplitudes are observed at $+3^\circ$ and -3° attack angle.

It can be seen that due to the limitation of the length of the wind fairing, triangular wind fairing cannot suppress the VIV of the railway rectangular steel box girder very well. The DOWNWARD quadrilateral wind fairing with a platform have better VIV suppression efficiency. M6 wind fairing is the optimal one and can eliminate VIVs of the railway

rectangular steel box girder. All of these consist well with the results of the numerical simulation.

Since coupled motions exist in the VIV response of the original section and the section with M0 wind fairing, the vertical and torsional amplitudes of them in Figs. 21 and 22 which are decoupled from the coupling vibration may be weakened (Wang and Zhang, 2021). Thus, the conclusion that some wind fairings even aggravate the VIV of the girder is only applicable to the torsional-bending frequency ratio of the Bianyuzhou Yangtze River Bridge in this paper. In order to compare the actual suppression efficiency of the triangular wind fairings, it is needed to completely separate different VIV ranges of the girder with or without wind fairings. It will be worth investigating. However, this does not affect the confirming of the suppression efficiency of the proposed DOWNWARD quadrilateral wind fairing.

6. Conclusion

In this paper, through a series of section model wind tunnel tests, the VIV performance of the high-speed railway cable-stayed bridge with a rectangular steel box girder is studied in detail. Then, the CFD method and dynamic mesh technology are employed to explore the triggering mechanism of the original rectangular steel box girder. According to the simulation results, four different size triangular wind fairings are designed first, and the flow pattern around the fixed girder installed with

the four triangular wind fairings are simulated through the CFD method. After that, a new type of wind fairing called DOWNWARD quadrilateral wind fairing with a platform is proposed, and the optimal wind fairing is found by the CFD method. Finally, the suppression efficiency of some typical wind fairings is verified through wind tunnel tests. Based on the test and simulation results, the following conclusions are made:

- (1) Severe VIVs are observed in the high-speed railway cable-stayed bridge with a rectangular steel box girder at 0° and $\pm 3^\circ$ attack angle. And VIVs of the main girder could not be eliminated by increasing damping ratios to about 0.5%.
- (2) For the original rectangular steel box girder with a breadth-to-depth ratio of 6.71, the leading-edge vortices and the after-body vortex shedding cause the deck VIVs. It should be noted that the leading-edge vortices on the upper and lower surfaces are the main cause.
- (3) θ_T and θ_B of 16° or 23° can eliminate the leading-edge vortices. Simply extending the length of the symmetrical triangular wind fairing to make both θ_T and θ_B reach 16° is uneconomic. A new type wind fairing called DOWNWARD quadrilateral wind fairing with a platform is proposed. This kind of wind fairing can not only eliminate the leading-edge vortex on the bottom surface of the girder, but also reduce that on the upper surface of the girder. The small upper leading-edge vortex is trapped because of the existence of the ballast wall. At $+3^\circ$ attack angle, the effect of the after-body vortex shedding on VIV is enhanced compared with that at 0° attack angle. Compared to the M4 and M5 wind fairing, the high frequency leading-edge vortex shedding and the weak after-body vortex shedding can suppress the VIVs at $+3^\circ$ attack angle after installing M6 fairing.
- (4) The suppression efficiency of the wind fairings obtained experimentally agrees well with that obtained in numerical simulations. For the VIVs of the high-speed railway cable-stayed bridge with a rectangular steel box girder, the suppression efficiency of triangular wind fairings is limited. The DOWNWARD quadrilateral wind fairing with a platform can suppress or even eliminate the VIVs, and the suppression efficiency is related to the length of the platform and the position of the nose.

CRedit authorship contribution statement

Jiahui Dong: Writing – original draft, Software, Visualization. **Lin Huang:** Data curation, Validation, Writing – review & editing. **Haili Liao:** Supervision, Methodology, Funding acquisition. **Qi Wang:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

National Natural Science Foundation of China under grant numbers 51778547 and 51678508 are gratefully acknowledged for the financial support of this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jweia.2021.104821>.

References

- Bartoli, G., Bruno, L., Buresti, G., Ricciarelli, F., Salvetti, M.V., Zasso, A., 2008a. BARC overview Document [WWW Document]. URL: <http://www.aniv-iawe.org/barcS>.
- Bartoli, G., Bruno, L., Buresti, G., Ricciarelli, F., Salvetti, M.V., Zasso, A., 2008b. Requests for computational simulations [WWW Document]. URL: <http://www.aniv-iawe.org/barcS>.
- Bruno, L., Coste, N., Fransos, D., 2012. Simulated flow around a rectangular 5:1 cylinder: spanwise discretisation effects and emerging flow features. *J. Wind Eng. Ind. Aerod.* 104–106, 203–215.
- De Miranda, M., Bartoli, G., 2001. Aerodynamic optimization of decks of cable-stayed bridges. In: IABSE Symposium, Seoul 2001: Cable-Supported Bridges - Challenging Technical Limits. International Association for Bridge and Structural Engineering (IABSE), Zurich, Switzerland, pp. 34–41.
- Hansen, S.O., 2013. Vortex-induced vibrations – the Scruton number revisited. *Proc. Inst. Civ. Eng. - Struct. Build.* 166, 560–571.
- Haque, M.N., Katsuchi, H., Yamada, H., Nishio, M., 2016. Investigation of edge fairing shaping effects on aerodynamic response of long-span bridge deck by unsteady RANS. *Arch. Civ. Mech. Eng.* 16, 888–900.
- Highway Agency of England, 2001. Design Manual for Roads and Bridges (Part 3) Design Rules for Aerodynamic Effects on Bridges. Report No. BD 49/01. Highway Agency of England, UK.
- Komatsu, S., Kobayashi, H., 1980. Vortex-induced oscillation of bluff cylinders. *J. Wind Eng. Ind. Aerod.* 6, 335–362.
- Kubo, Y., Honda, K., Tasaki, K., Kato, K., 1993. Improvement of aerodynamic instability of cable-stayed bridge deck by separated flows mutual interference method. *J. Wind Eng. Ind. Aerod.* 49, 553–564.
- Larsen, A., Wall, A., 2012. Shaping of bridge box girders to avoid vortex shedding response. *J. Wind Eng. Ind. Aerod.* 104–106, 159–165.
- Li, Y., Chen, K., Wang, B., Sun, H., 2018a. Optimal measures for vortex-induced vibration of a bluff girder with separated twin-box. *Zhendong yu Chongji/Journal Vib. Shock* 37, 116–122.
- Li, Y., Le, Chen, X.Y., Yu, C.J., Togbenou, K., Wang, B., Zhu, L.D., 2018b. Effects of wind fairing angle on aerodynamic characteristics and dynamic responses of a streamlined trapezoidal box girder. *J. Wind Eng. Ind. Aerod.* 177, 69–78.
- Li, M., Sun, Y., Li, Mingshui, Wu, B., 2018c. Vortex-induced vibration performance of wide streamlined box girder and aerodynamic countermeasure research. *Xinan Jiaotong Daxue Xuebao/Journal Southwest Jiaotong Univ.* 53, 712–719.
- Ma, C.M., Wang, J.X., Li, Q.S., Qin, H., Liao, H.L., 2018. Vortex-induced vibration performance and suppression mechanism for a long suspension bridge with wide twin-box girder. *J. Struct. Eng.* 144, 04018202.
- Matsumoto, M., Shirato, H., Araki, K., Haramura, T., Hashimoto, T., 2003. Spanwise coherence characteristics of surface pressure field on 2-D bluff bodies. *J. Wind Eng. Ind. Aerod.* 91, 155–163.
- McLean, L., Gartshore, I., 1992. Spanwise correlations of pressure on a rigid square section cylinder. *J. Wind Eng. Ind. Aerod.* 41, 797–808.
- Meng, X.L., Guo, Z.S., Ding, Q.S., Zhu, L.D., 2011. Influence of wind fairing angle on vortex-induced vibrations and flutter performances of closed and semi-closed box decks. *Gongcheng Lixue/Engineering Mech.* 28, 6–11.
- Ministry of Transport of the People's Republic of China, 2018. Wind-resistant Design Specification for Highway Bridges. Report No. JTG/T 3360-01-2018. China Communications Press, Beijing, China.
- Murakami, S., Mochida, A., 1995. On turbulent vortex shedding flow past 2D square cylinder predicted by CFD. *J. Wind Eng. Ind. Aerod.* 54–55, 191–211.
- Nagao, F., Utsunomiya, H., Oryu, T., Manabe, S., 1993. Aerodynamic efficiency of triangular fairing on box girder bridge. *J. Wind Eng. Ind. Aerod.* 49, 565–574.
- Nakamura, Y., Nakashima, M., 1986. Vortex excitation of prisms with elongated rectangular, H and T cross-sections. *J. Fluid Mech.* 163, 149–169.
- Namirani, F., Gartshore, I.S., 1988. Direct measurements of oscillating lift on a rigid square section cylinder in a turbulent stream. *J. Wind Eng. Ind. Aerod.* 28, 209–218.
- Norberg, C., 1993. Flow around rectangular cylinders: pressure forces and wake frequencies. *J. Wind Eng. Ind. Aerod.* 49, 187–196.
- Sato, H., 2003. Wind-resistant design manual for highway bridges in Japan. *J. Wind Eng. Ind. Aerod.* 91, 1499–1509.
- Shimada, K., Ishihara, T., 2002. Application of a modified k- ϵ model to the prediction of aerodynamic characteristics of rectangular cross-section cylinders. *J. Fluid Struct.* 16, 465–485.
- Shiraishi, N., Matsumoto, M., 1983. On classification of vortex-induced oscillation and its application for bridge structures. *J. Wind Eng. Ind. Aerod.* 14, 419–430.
- Sohankar, A., Norberg, C., Davidson, L., 1999. Simulation of three-dimensional flow around a square cylinder at moderate Reynolds numbers. *Phys. Fluids* 11, 288–306.
- Srinivas, Y., Biswas, G., Parihar, A.S., Ranjan, R., 2006. Large-eddy simulation of high Reynolds number turbulent flow past a square cylinder. *J. Eng. Mech.* 132, 327–335.
- Sukanta Nagao, F., Noda, M., Muneta, K., 2008. Aerodynamic stabilizing mechanism of a cable stayed bridge with two edge box girder. In: Proc. Of 6th Int. Colloquium on Bluff Body Aerodynamics and Applications, pp. 20–24.
- Sun, D., Owen, J.S., Wright, N.G., 2009. Application of the k- ω turbulence model for a wind-induced vibration study of 2D bluff bodies. *J. Wind Eng. Ind. Aerod.* 97, 77–87.
- Vaz, D.C., Almeida, R.A.B., Didier, E., Uguieira, A.P.V., Borges, A.R.J., 2016. Improving the aerodynamic performance of Vila-Real Bridge deck-section. *J. Wind Eng. Ind. Aerod.* 156, 72–83.
- Wang, Q., Liao, H., Li, M., Ma, C., 2011. Influence of aerodynamic configuration of a streamline box girder on bridge flutter and vortex-induced vibration. *J. Mod. Transp.* 19, 261–267.
- Wang, Z., Zhang, Z., 2021. VIV properties of π -shaped bridge sectional model: dependence on torsional-bending frequency ratio. *J. Bridge Eng.* 26, 6021003.

- Xu, K., Ge, Y., Zhao, L., Du, X., 2018. Experimental and numerical study on the dynamic stability of vortex-induced vibration of bridge decks. *Int. J. Struct. Stabil. Dynam.* 18, 1–15.
- Yu, D., Kareem, A., 1998. Parametric study of flow around rectangular prisms using LES. *J. Wind Eng. Ind. Aerod.* 77–78, 653–662.
- Zhang, T., Sun, Y., Li, M., Yang, X., 2020. Experimental and numerical studies on the vortex-induced vibration of two-box edge girder for cable-stayed bridges. *J. Wind Eng. Ind. Aerod.* 206, 104336.
- Zhou, R., Ge, Y., Yang, Y., Du, Y., Zhang, L., 2020. Aerodynamic performance evaluation of different cable-stayed bridges with composite decks. *Steel Compos. Struct.* 34, 699–713.
- Zhou, R., Ge, Y., Yang, Y., Du, Y., Zhang, L., 2018. Wind-induced nonlinear behaviors of twin-box girder bridges with various aerodynamic shapes. *Nonlinear Dynam.* 94, 1095–1115.